

Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

Meeting 11 OllamaModelTester Web – GitHub CI/CD Pipeline
Automation

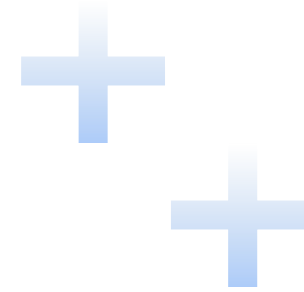
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Agenda

- Problem Statement
- Solution Overview
- Architecture Diagram
- Workflow Trigger & Runner
- Secure Remote Access with Secrets
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- Steps: Rollout Restart & Verification
- Steps: Cleanup (Housekeeping)



3 Problem Statement



Why automate deployment?

- Manual SSH/RDP + Docker build + kubectl apply = slow and error-prone
- Drift between what's in Git and what's running
- No audit trail of who deployed what
- Long feedback loop for developers
- Goal: One git push → fully deployed, verified, and cleaned-up environment

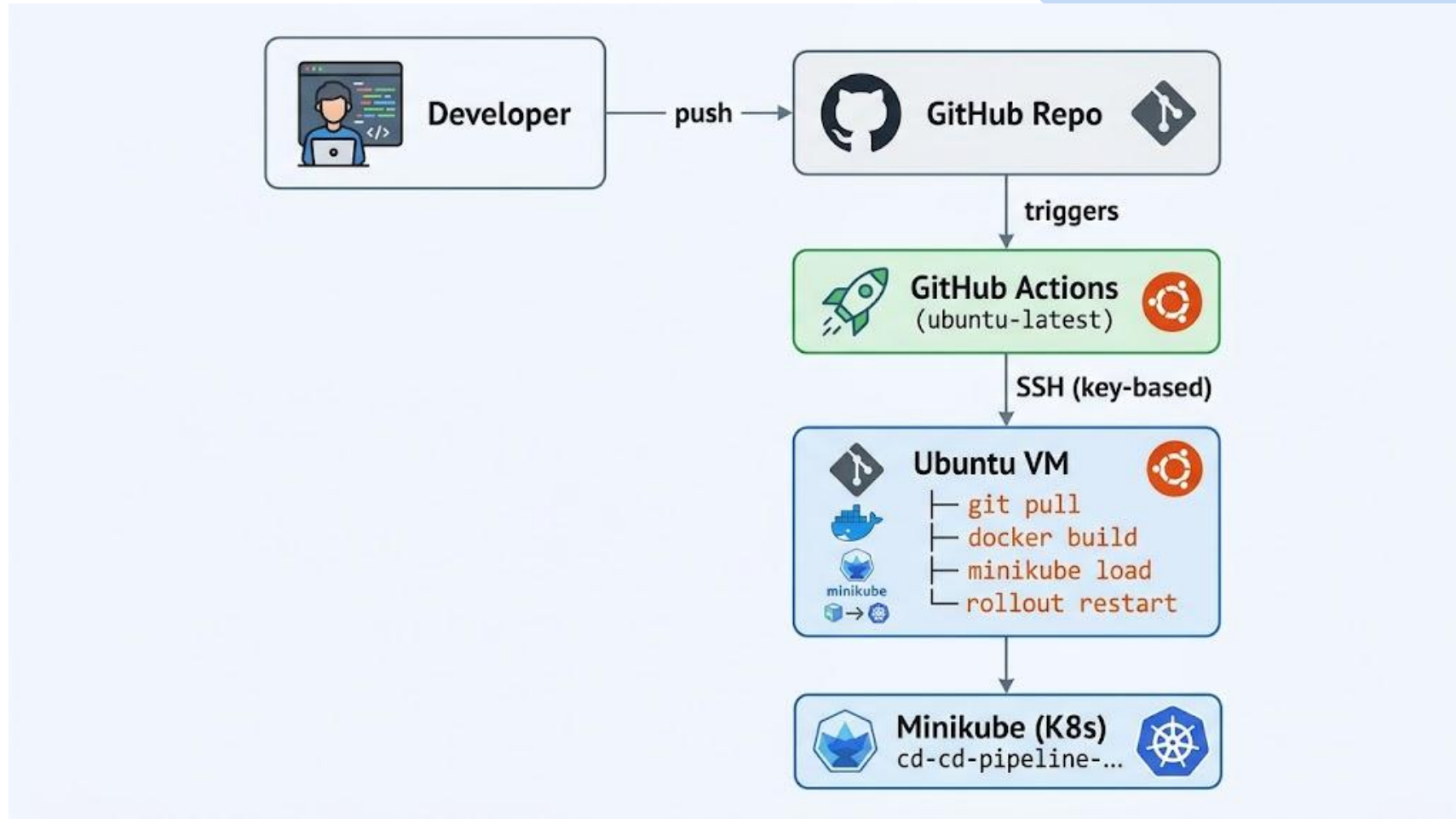
4 Solution Overview

GitHub Actions → SSH → Ubuntu VM → Minikube

- Trigger: push to main
- Runner: ubuntu-latest (GitHub-hosted)
- Remote execution via appleboy/ssh-action@v1.2.5
- Target: Ubuntu VM running Kubernetes (Minikube)
- Outcome: Fresh image built, loaded into cluster, deployment restarted



Architecture Diagram



Workflow Trigger & Runner

- Trigger: every push to main (extendable to PRs, tags, manual workflow dispatch)
- Runner: GitHub-hosted Ubuntu – no self-hosted overhead
- Isolation: each run is a clean environment

```
name: Deploy to Ubuntu VM and Kubernetes
```

```
on:
```

```
  push:
```

```
    branches: [ "main" ]
```

```
jobs:
```

```
  deploy:
```

```
    runs-on: ubuntu-latest
```

Secure Remote Access with Secrets

- Credentials stored in GitHub Encrypted Secrets – never in code
- `script_stop: true` → fail fast on any command error
- `command_timeout: 30m` → headroom for image builds

```
with:  
  host: ${ secrets.VM_HOST }  
  username: ${ secrets.VM_USER }  
  key: ${ secrets.VM_SSH_KEY }  
  command_timeout: 30m  
  script_stop: true
```

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Steps: Checkout & Pull Latest Code

- Fixed working directory with fail-safe or exit 1
- Ensures VM manifest + source always match main
- git stash option available (commented) for local-change safety

```
# 1. Navigate to the application directory
cd /home/ubuntu/.projects/[Project Folder]/ || exit 1

# 2. Pull the latest changes
# Stash any local changes to avoid conflicts
git pull origin main
```

Steps: Build & Load Image into Kubernetes (Minikube)

- Image built on the VM – no registry needed
- minikube image load pushes the image directly into the cluster's runtime

```
# Create a Docker image in Kubernetes (Minikube)
minikube image build -t ollama-tester-backend:latest ./backend/.

# Publish the deployment and service in Kubernetes (Minikube)
minikube kubectl -- apply -f ./backend/k8s/backend-deployment.yaml
```

Steps: Rollout Restart & Verification

- rollout restart forces pods to pick up the new :latest image
- rollout status blocks until success or fails the pipeline on timeout
- Guarantees the workflow only reports success on a healthy deployment

```
# Restart the Deployment
```

```
minikube kubectl -- rollout restart deployment/ollama-tester-backend:latest
```

```
# Verify the rollout
```

```
minikube kubectl -- rollout status deployment/ollama-tester-backend:latest --timeout=90s
```

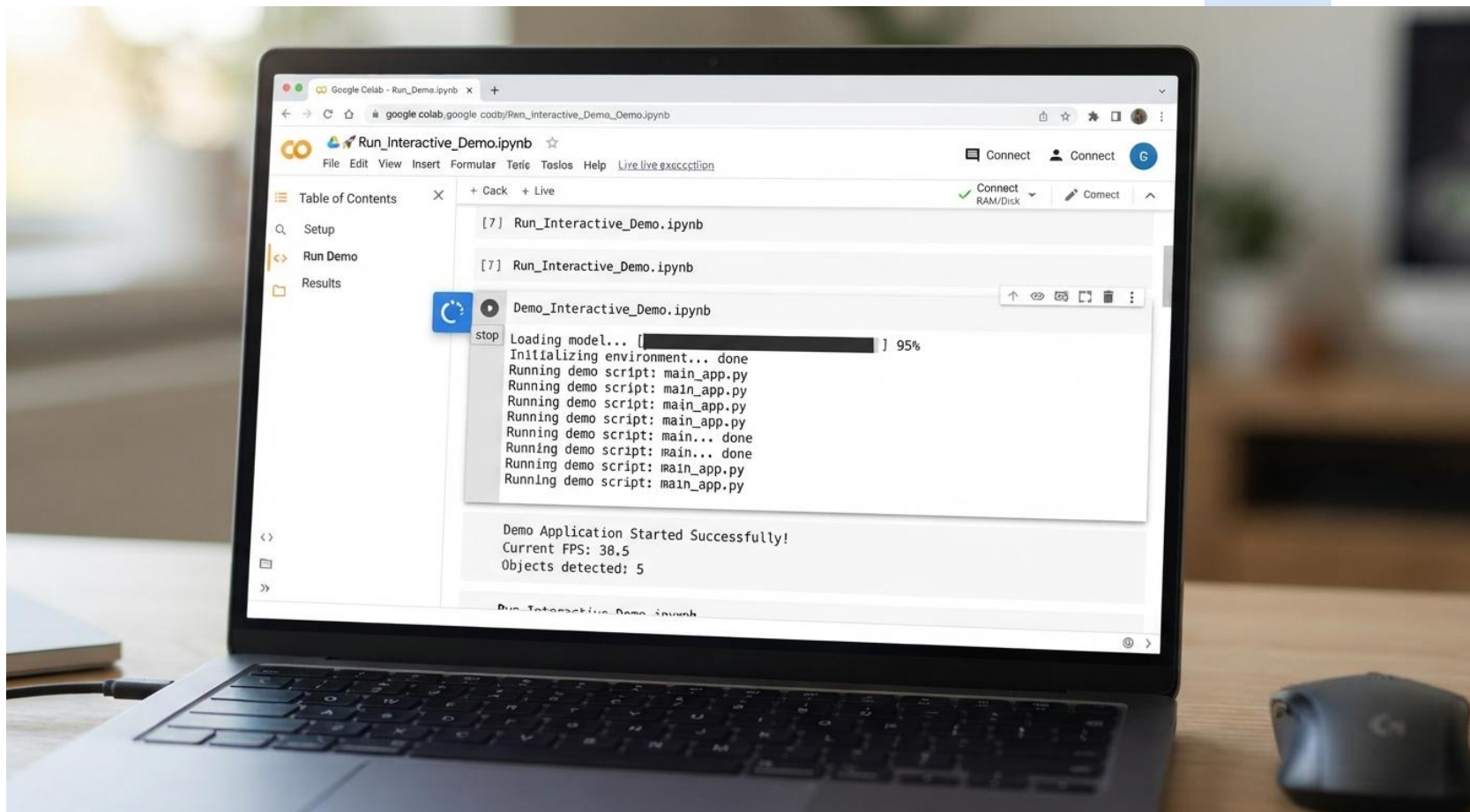
Steps: Cleanup (Housekeeping)

- Removes unused images, stopped containers, unused networks
- Clears the BuildKit cache to reclaim disk
- Prevents VM disk exhaustion on long-running pipelines

```
docker system prune -a -f  
docker builder prune -a -f
```

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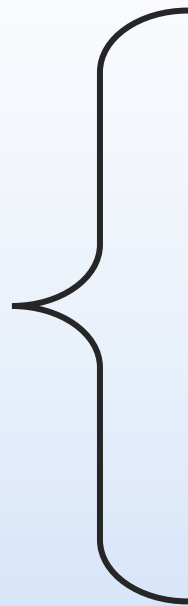
LIVE DEMO



References

- ed25519 signature algorithm – <https://messari.io/copilot/share/understanding-ed25519-ee5c6e19-3f05-4557-ac8f-9a1e84b9e8ff>
- GitHub on Push to Main – <https://docs.github.com/en/actions/reference/workflows-and-actions/workflow-syntax>
- Actions/Checkout – <https://github.com/actions/checkout>
- Appleboy SSH for GitHub Actions – <https://github.com/appleboy/ssh-action>
- Setting CI/CD pipeline using GitHub Actions in GCP – <https://github.com/orgs/community/discussions/161973>

Thank you



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